

Analiza Algoritmilor

SAT and a few polynomial reductions

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Boolean sentences

Start with a set of propositions

`Props = {p, q, r, … , x, y, z, …}`

Sometimes they are called variables.
It is not a mistake but it is awkward.

Boolean sentences

Start with a set of propositions

$\mathtt{Props} = \{p, q, r, \ldots, x, y, z, \ldots\}$

A proposition can be true or false.

An interpretation tells us which propositions are true and which are false.

$\mathtt{I} : \mathtt{Props} \rightarrow \{0,1\}$ or

$\mathtt{I} \subseteq \mathtt{Props}$

Propositions in $\mathtt{I}$ are held to be true and the rest – false.

Boolean sentences

A sentence φ is a proposition ($\varphi = p$)

or a conjunction ($\varphi = \varphi_1 \wedge \varphi_2$)

or a disjunction ($\varphi = \varphi_1 \vee \varphi_2$)

or a negation ($\varphi = \neg\varphi_1$)

or parentheses ($\varphi = (\varphi_1)$)

parentheses constructors per se, but they help us disambiguate propositions just like in arithmetic

Boolean sentences

The sentence $\varphi_1 \rightarrow \varphi_2$ means $\neg\varphi_1 \vee \varphi_2$

The sentence $\varphi_1 \leftrightarrow \varphi_2$ means $(\varphi_1 \rightarrow \varphi_2) \wedge (\varphi_2 \rightarrow \varphi_1)$

Conjunctive Normal Form (CNF)

The sentence φ is in CNF iff

$\varphi = C_1 \wedge C_2 \wedge \ldots \wedge C_n$ where

$C_i = L_{i1} \wedge L_{i2} \wedge \ldots \wedge L_{ik}$ where

$L_{ij} = p$ or $\neg p$ where p is any proposition

Conjunctive Normal Form (CNF) vs arbitrary form

$\varphi = (p \vee q \wedge \neg r) \wedge \neg(p \wedge (q \vee \neg r))$

Representing a sentence requires a tree

Conjunctive Normal Form (CNF) vs arbitrary form

$$\varphi = (p \vee q \wedge \neg r) \wedge \neg(p \wedge (q \vee \neg r))$$

We can transform this sentence in CNF

Conjunctive Normal Form (CNF) vs arbitrary form

$$\varphi = (p \vee q) \wedge (p \vee \neg r) \wedge \neg(p \wedge (q \vee \neg r))$$

Conjunctive Normal Form (CNF) vs arbitrary form

$$\varphi = (p \vee q) \wedge (p \vee \neg r) \wedge (\neg p \vee \neg(q \vee \neg r))$$

Conjunctive Normal Form (CNF) vs arbitrary form

$$\varphi = (p \vee q) \wedge (p \vee \neg r) \wedge (\neg p \vee \neg q \wedge r)))$$

Conjunctive Normal Form (CNF) vs arbitrary form

$$\varphi = (p \vee q) \wedge (p \vee \neg r) \wedge (\neg p \vee \neg q) \wedge (\neg p \vee r)$$

Conjunctive Normal Form (CNF) vs arbitrary form

$$\varphi = (p \vee q) \wedge (p \vee \neg r) \wedge (\neg p \vee \neg q) \wedge (\neg p \vee r)$$

Conjunctive Normal Form (CNF) vs arbitrary form

$$\varphi = (p \vee q) \\ (p \vee \neg r) \\ (\neg p \vee \neg q) \\ (\neg p \vee r)$$

each line is a conjunction

Conjunctive Normal Form (CNF) vs arbitrary form

$$
\begin{aligned}
\varphi = (\ & p \vee q \quad\ \ ) \\
(\ & p \vee \quad\ \ \neg r) \\
(\neg & p \vee \neg q \quad\ ) \\
(\neg & p \vee \quad\ \ r)
\end{aligned}
$$

each line is a conjunction

Conjunctive Normal Form (CNF) vs arbitrary form

$$\varphi = \begin{matrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \\ -1 & 0 & 1 \end{matrix}$$

each element L_{ij} tells us if proposition j appears positive, negated, or not at all, in clause i

Boolean sentences

A sentence is called **satisfiable** iff there **exists** an interpretation (for each proposition of the formula), which makes the formula true.

Is this formula **satisfiable**?

$$\varphi = (p \vee q \wedge \neg r) \wedge \neg(p \wedge (q \vee \neg r))$$

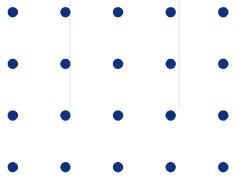
Hard to ***eyeball*** the answer



Boolean sentences

Is this formula **satisfiable**?

$$\varphi = (p \vee q) \wedge (p \vee \neg r) \wedge (\neg p \vee \neg q) \wedge (\neg p \vee r)$$



Boolean sentences

Is this formula satisfiable?

$$\varphi = (p \vee q) \wedge (p \vee \neg r) \wedge (\neg p \vee \neg q) \wedge (\neg p \vee r)$$

We search for an interpretation which makes all clauses true

In each clause at least one literal must be true

SAT:

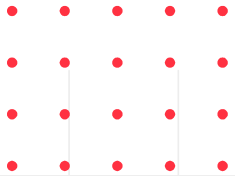
Given a sentence in CNF, is it satisfiable?

SAT ∈ *NP*

```
SAT-algorithm(phi, n, m):
I = array of size m.

for p = 1,m:
    I[p] = choice({-1,1})

for i = 1,n:
    current_clause = false
    for l = 1,m:
        if phi[i][j]*I[j] > 0:
            current_clause = true
    if !current_clause:
        fail
success
```



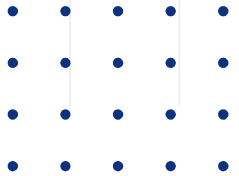
SAT:

Given a sentence in CNF, is it satisfiable?

SAT is ***NP-complete***

Cook-Levine Theorem (1970)





How is SAT useful to us?

usage 1

We want to solve a difficult problem f, for which we do not know an algorithm:

- We find a transformation witnessing $f \leq_p SAT$
  We transform the input to our problem in a SAT formula and run a SAT solver.

usage 2

We want to prove that a problem f, is NP-hard.

- We find a transformation witnessing $SAT \leq_p f$
  - We can also choose other NP-hard problems, not necessarily SAT.

We want to solve a difficult problem **f**, for which we do not know an algorithm

Linear programming

Is a set of inequalities over integers satisfiable?

$$T_{11} + T_{12} + \dots + T_{1n1} \leq c_1$$
$$T_{21} + T_{22} + \dots + T_{2n2} \leq c_2$$
$$\dots$$
$$T_{k1} + T_{k2} + \dots + T_{knk} \leq c_k$$

We want to solve a difficult problem **f**, for which we do not know an algorithm

Linear programming

Is a set of inequalities over integers satisfiable?

$x_1 + x_2 + x_3 + x_4 \leq 1$ *at most one variable is 1*

$x_1 + x_2 + x_3 = 1$ *exactly one variable is 1*

$$(x_1 \wedge \neg x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)$$

not in CNF

$$(x_1 \wedge \neg x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)$$

$(x_1 \wedge \neg x_2 \wedge \neg x_3) \vee \text{rest}$

$$(x_1 \vee \text{rest}) \wedge (\neg x_2 \vee \text{rest}) \wedge (\neg x_3 \vee \text{rest})$$

$(x_1 \vee \text{rest}) \wedge$
$(\neg x_2 \vee \text{rest}) \wedge$
$(\neg x_3 \vee \text{rest})$

$$(x_1 \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge$$
$$(\neg x_2 \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge$$
$$(\neg x_3 \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge \neg x_2 \wedge x_3))$$

$$(x_1 \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge$$
$$(\neg x_2 \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge$$
$$(\neg x_3 \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge \neg x_2 \wedge x_3))$$

$$(x_1 \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee \text{rest}) \wedge$$
$$(\neg x_2 \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee \text{rest}) \wedge$$
$$(\neg x_3 \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee \text{rest})$$

$$((x_1 \vee \neg x_1 \vee \text{rest}) \wedge (x_1 \vee x_2 \wedge \text{rest}) \wedge (x_1 \vee \neg x_3 \wedge \text{rest})) \wedge$$
$$((\neg x_2 \vee \neg x_1 \vee \text{rest}) \wedge (\neg x_2 \vee x_2 \vee \text{rest}) \wedge (\neg x_2 \vee \neg x_3 \vee \text{rest})) \wedge$$
$$((\neg x_3 \vee \neg x_1 \vee \text{rest}) \wedge (\neg x_3 \vee x_2 \vee \text{rest}) \wedge (\neg x_3 \vee \neg x_3 \vee \text{rest}))$$

$$
\begin{aligned}
&((x_1 \vee \neg x_1 \vee \text{rest}) \wedge \\
&(x_1 \vee x_2 \vee \text{rest}) \wedge \\
&(x_1 \vee \neg x_3 \vee \text{rest})) \wedge \\
&((\neg x_2 \vee \neg x_1 \vee \text{rest}) \wedge \\
&(\neg x_2 \vee x_2 \vee \text{rest}) \wedge \\
&(\neg x_2 \vee \neg x_3 \vee \text{rest})) \wedge \\
&((\neg x_3 \vee \neg x_1 \vee \text{rest}) \wedge \\
&(\neg x_3 \vee x_2 \vee \text{rest}) \wedge \\
&(\neg x_3 \vee \neg x_3 \vee \text{rest}))
\end{aligned}
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$$
\begin{aligned}
&((x_1 \vee \neg x_1 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge \\
&(x_1 \vee x_2 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge \\
&(x_1 \vee \neg x_3 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3))) \wedge \\
&((\neg x_2 \vee \neg x_1 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge \\
&(\neg x_2 \vee x_2 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge \\
&(\neg x_2 \vee \neg x_3 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3))) \wedge \\
&((\neg x_3 \vee \neg x_1 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge \\
&(\neg x_3 \vee x_2 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge \\
&(\neg x_3 \vee \neg x_3 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)))
\end{aligned}
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$$((x_1 \vee \neg x_1 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge$$
$$(x_1 \vee x_2 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge$$
$$(x_1 \vee \neg x_3 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3))) \wedge$$
$$((\neg x_2 \vee \neg x_1 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge$$
$$(\neg x_2 \vee x_2 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge$$
$$(\neg x_2 \vee \neg x_3 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3))) \wedge$$
$$((\neg x_3 \vee \neg x_1 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge$$
$$(\neg x_3 \vee x_2 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)) \wedge$$
$$(\neg x_3 \vee \neg x_3 \vee (\neg x_1 \wedge \neg x_2 \wedge x_3)))$$

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\begin{aligned}
&((x_1 \vee \neg x_1 \vee \neg x_1) \wedge (x_1 \vee \neg x_1 \vee \neg x_2) \wedge (x_1 \vee \neg x_1 \vee x_3)) \wedge \\
&((x_1 \vee x_2 \vee \neg x_1) \wedge (x_1 \vee x_2 \vee \neg x_2) \wedge (x_3 \vee x_1 \vee x_2)) \wedge \\
&((x_1 \vee \neg x_3 \vee \neg x_1) \wedge (x_1 \vee \neg x_3 \vee \neg x_2) \wedge (x_1 \vee \neg x_3 \vee x_3)) \wedge \\
&((\neg x_2 \vee \neg x_1 \vee \neg x_1) \wedge (\neg x_2 \vee \neg x_1 \vee \neg x_2) \wedge (\neg x_2 \vee \neg x_1 \vee x_3)) \wedge \\
&((\neg x_2 \vee x_2 \vee \neg x_1) \wedge (\neg x_2 \vee x_2 \vee \neg x_2) \wedge (\neg x_2 \vee x_2 \vee x_3)) \wedge \\
&((\neg x_2 \vee \neg x_3 \vee \neg x_1) \wedge (\neg x_2 \vee \neg x_3 \vee \neg x_2) \wedge (\neg x_2 \vee \neg x_3 \vee x_3)) \wedge \\
&((\neg x_3 \vee \neg x_1 \vee \neg x_1) \wedge (\neg x_3 \vee \neg x_1 \vee \neg x_2) \wedge (\neg x_3 \vee \neg x_1 \vee x_3)) \wedge \\
&((\neg x_3 \vee x_2 \vee \neg x_1) \wedge (\neg x_3 \vee x_2 \vee \neg x_2) \wedge (\neg x_3 \vee x_2 \vee x_3)) \wedge \\
&((\neg x_3 \vee \neg x_3 \vee \neg x_1) \wedge (\neg x_3 \vee \neg x_3 \vee \neg x_2) \wedge (\neg x_3 \vee \neg x_3 \vee x_3)))
\end{aligned}
$$

$$(x_1 \vee \neg x_1 \vee \neg x_1) \wedge (x_1 \vee \neg x_1 \vee \neg x_2) \wedge (x_1 \vee \neg x_1 \vee x_3) \wedge$$
$$(x_1 \vee x_2 \vee \neg x_1) \wedge (x_1 \vee x_2 \vee \neg x_2) \wedge (x_3 \vee x_1 \vee x_2) \wedge$$
$$(x_1 \vee \neg x_3 \vee \neg x_1) \wedge (x_1 \vee \neg x_3 \vee \neg x_2) \wedge (x_1 \vee \neg x_3 \vee x_3) \wedge$$
$$(\neg x_2 \vee \neg x_1 \vee \neg x_1) \wedge (\neg x_2 \vee \neg x_1 \vee \neg x_2) \wedge (\neg x_2 \vee \neg x_1 \vee x_3) \wedge$$
$$(\neg x_2 \vee x_2 \vee \neg x_1) \wedge (\neg x_2 \vee x_2 \vee \neg x_2) \wedge (\neg x_2 \vee x_2 \vee x_3) \wedge$$
$$(\neg x_2 \vee \neg x_3 \vee \neg x_1) \wedge (\neg x_2 \vee \neg x_3 \vee \neg x_2) \wedge (\neg x_2 \vee \neg x_3 \vee x_3) \wedge$$
$$(\neg x_3 \vee \neg x_1 \vee \neg x_1) \wedge (\neg x_3 \vee \neg x_1 \vee \neg x_2) \wedge (\neg x_3 \vee \neg x_1 \vee x_3) \wedge$$
$$(\neg x_3 \vee x_2 \vee \neg x_1) \wedge (\neg x_3 \vee x_2 \vee \neg x_2) \wedge (\neg x_3 \vee x_2 \vee x_3) \wedge$$
$$(\neg x_3 \vee \neg x_3 \vee \neg x_1) \wedge (\neg x_3 \vee \neg x_3 \vee \neg x_2) \wedge (\neg x_3 \vee \neg x_3 \vee x_3)$$

Original formula:

$$(x_1 \wedge \neg x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee (\neg x_1 \wedge \neg x_2 \wedge x_3).$$

CNF transformation is **exponential in size** in the worst case

We want to solve a difficult problem **f**, for which we do not know an algorithm

Linear programming

Is a set of inequalities over integers satisfiable?

$x_1 + x_2 + x_3 + x_4 \leq 1$ *at most one variable is 1*

$(\neg x_1 \vee \neg x_2) \wedge$ *one of* x_1, x_2 *is* ***zero****, and*
$(\neg x_1 \vee \neg x_3) \wedge$ *one of* x_1, x_3 *is* ***zero****, and*
$(\neg x_1 \vee \neg x_4) \wedge$ *one of* x_1, x_4 *is* ***zero****, and*
$(\neg x_2 \vee \neg x_3) \wedge$ *one of* x_2, x_3 *is* ***zero*** *, and*
$(\neg x_2 \vee \neg x_4) \wedge$ *one of* x_2, x_4 *is* ***zero*** *, and*
$(\neg x_3 \vee \neg x_4)$ *one of* x_3, x_4 *is* ***zero*** *.*

$O(n^2)$ literals for n variables

We want to solve a difficult problem f, for which we do not know an algorithm

Linear programming

Is a set of inequalities over integers satisfiable?

$$x_1 + x_2 + x_3 + x_4 \leq 1$$

at most one variable is 1

$$\text{atMostOne}(x_1, x_2, \ldots, x_n) = \bigwedge_{i \neq j} (\neg x_i \vee \neg x_j)$$

We want to solve a difficult problem **f**, for which we do not know an algorithm

Linear programming

Is a set of inequalities over integers satisfiable?

$x_1 + x_2 + x_3 + x_4 \leq 1$ *at most one variable is 1*

$x_1 + x_2 + x_3 = 1$ *exactly one variable is 1*

We want to solve a difficult problem f, for which we do not know an algorithm

Linear programming

Is a set of inequalities over integers satisfiable?

$$x_1 + x_2 + x_3 + x_4 \leq 1$$

at most one variable is 1

$$x_1 + x_2 + x_3 = 1$$

$$\text{exactlyOne}(x_1, x_2, \ldots, x_n) =$$

$$\text{atMostOne}(x_1, x_2, \ldots, x_n) \wedge$$

$$\text{atLeastOne}(x_1, x_2, \ldots, x_n)$$

We want to solve a difficult problem f, for which we do not know an algorithm

Linear programming

Is a set of inequalities over integers satisfiable?

$$x_1 + x_2 + x_3 + x_4 \leq 1$$

$$x_1 + x_2 + x_3 = 1$$

at most one variable is 1

$$\text{exactlyOne}(x_1, x_2, \ldots, x_n) =$$

$$\text{atMostOne}(x_1, x_2, \ldots, x_n) \wedge$$

$$(x_1 \vee x_2 \vee \ldots \vee x_n)$$

This set of inequalities is satisfiable

$$x_1 + x_2 + x_3 + x_4 \leq 1$$

$$x_1 + x_2 + x_3 = 1$$

if and only if the following CNF formula is satisfiable

$$(\neg x_1 \vee \neg x_2) \wedge (\neg x_1 \vee \neg x_3) \wedge (\neg x_1 \vee \neg x_4) \wedge (\neg x_2 \vee \neg x_3) \wedge (\neg x_2 \vee \neg x_4) \wedge (\neg x_3 \vee \neg x_4) \wedge (\neg x_1 \vee \neg x_2) \wedge (\neg x_1 \vee \neg x_3) \wedge (\neg x_2 \vee \neg x_3) \wedge (x_1 \vee x_2 \vee x_3)$$

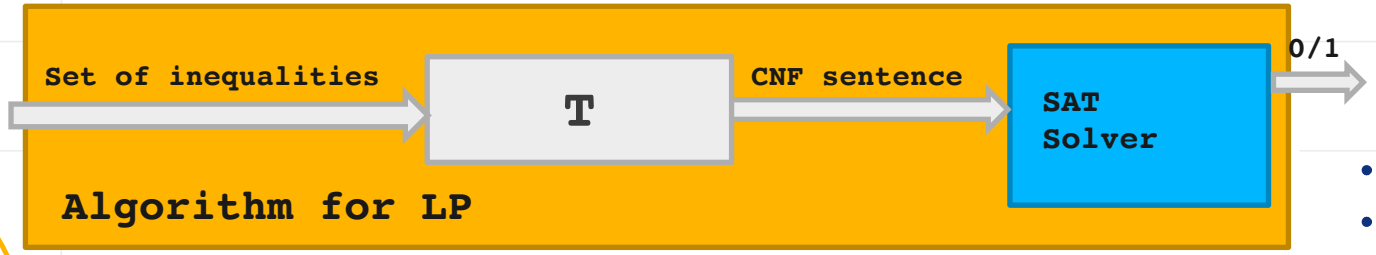
We also know of encodings for more general equations where c > 1

$$T_{11} + T_{12} + \ldots + T_{1n1} \leq c_1$$

They are more elaborate and go beyond the scope of this lecture

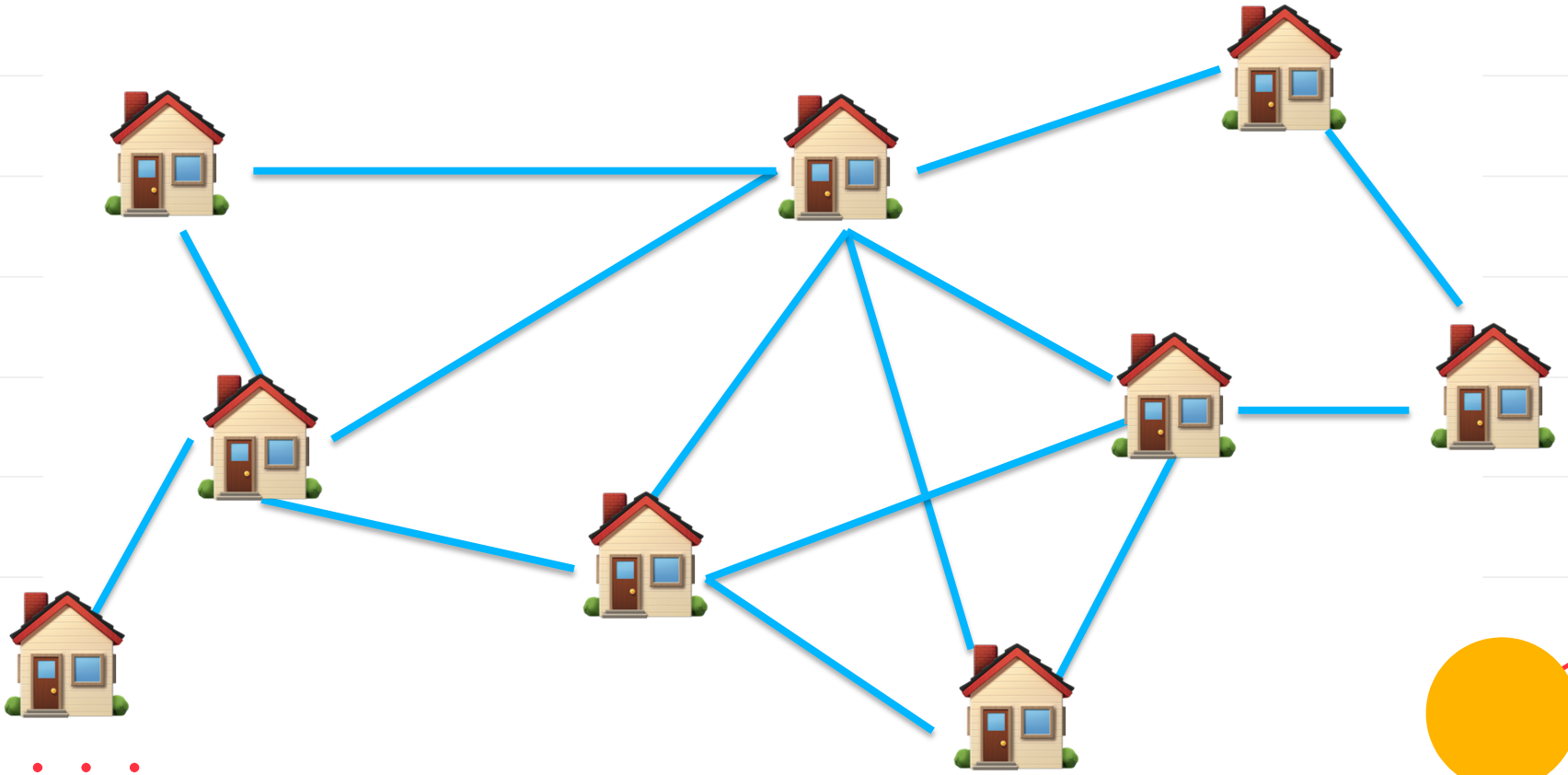
We must prove that the inequalities are satisfiable *iff* the CNF formula we have built is satisfiable.

$$T_{11} + T_{12} + \dots + T_{1n1} \leq c_1$$
$$T_{21} + T_{22} + \dots + T_{2n2} \leq c_2$$
$$\dots$$
$$T_{k1} + T_{k2} + \dots + T_{knk} \leq c_k$$

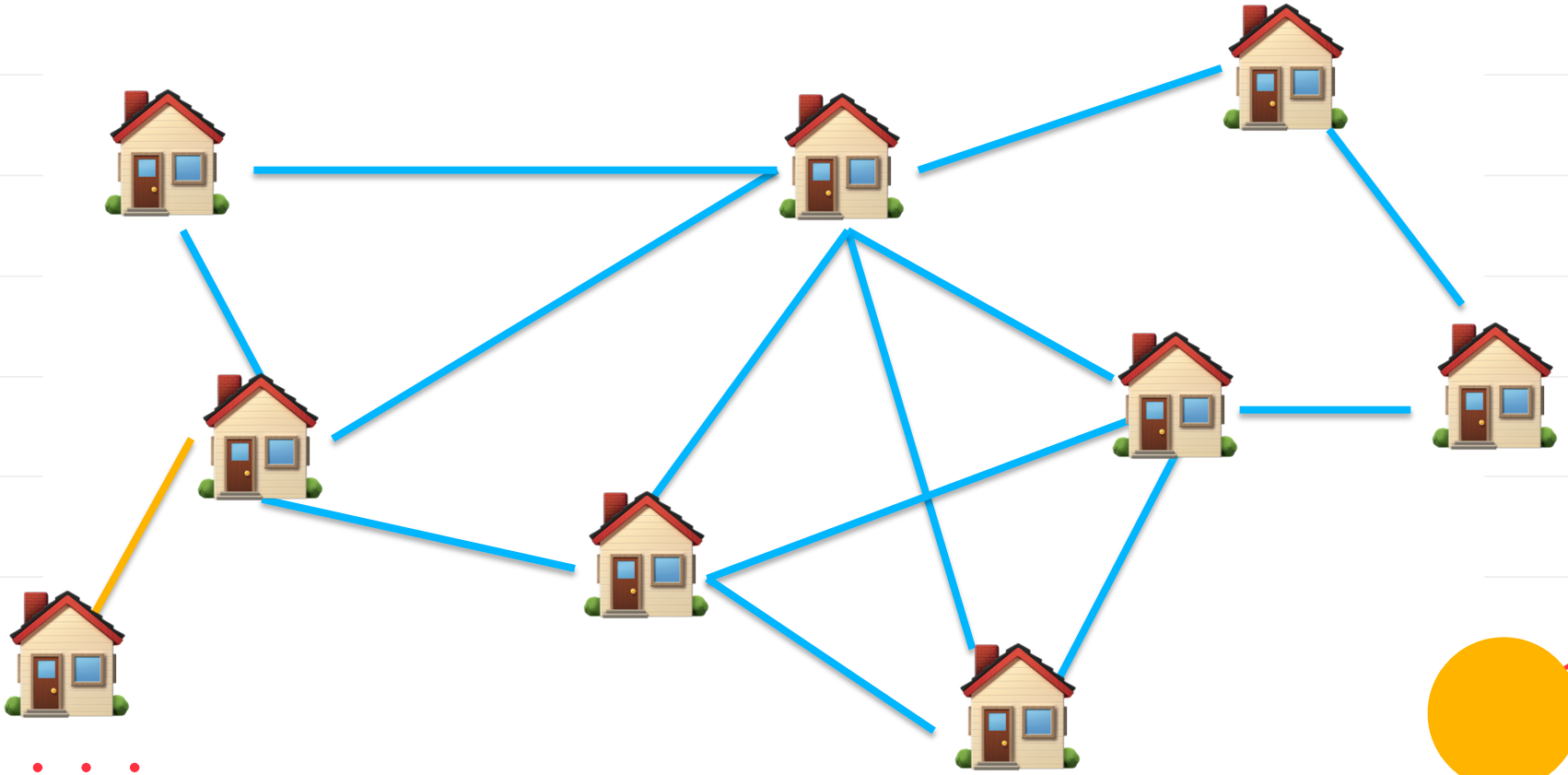


$LP \leq_p SAT$

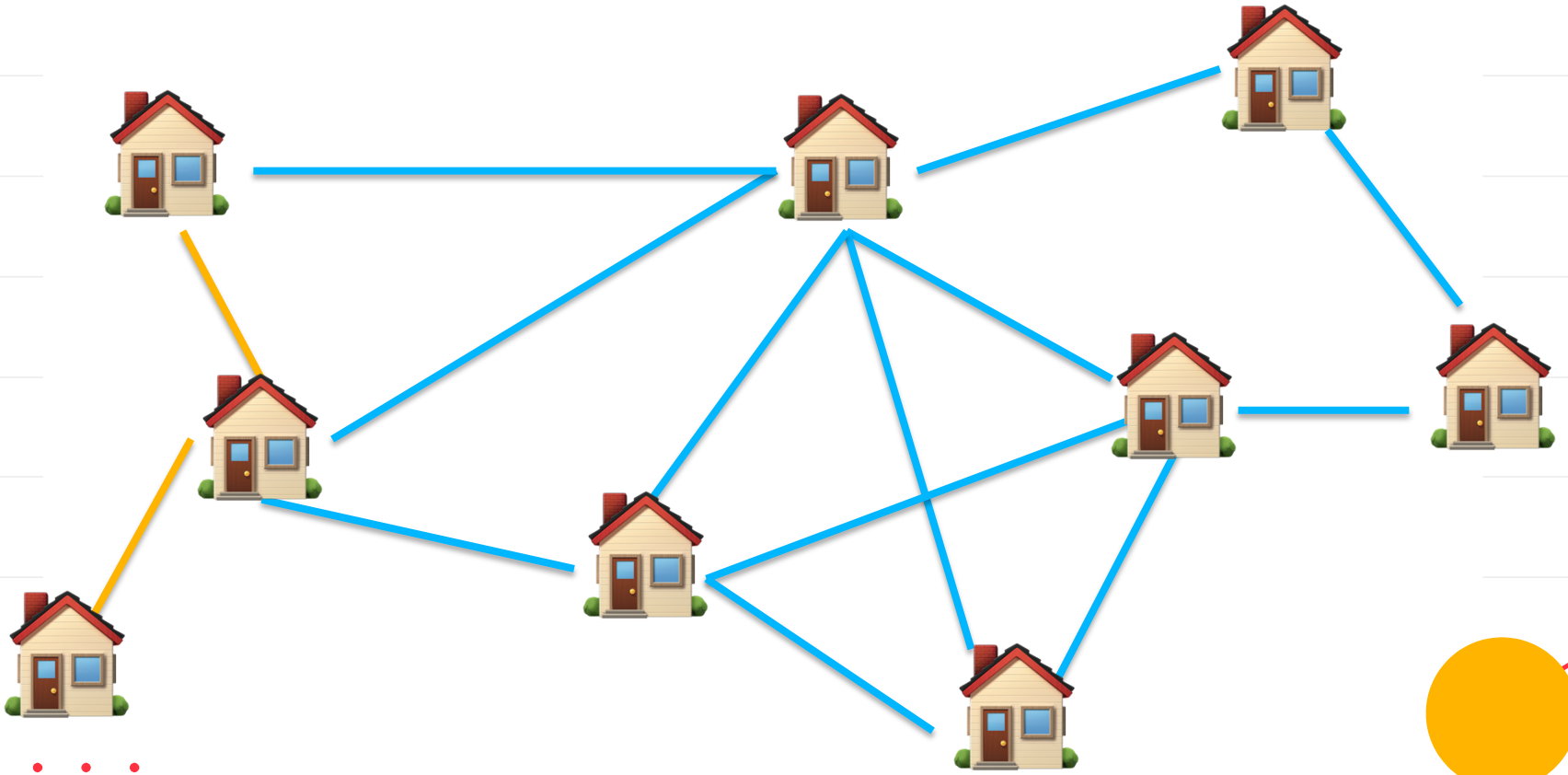
Practice: Hamiltonian Path



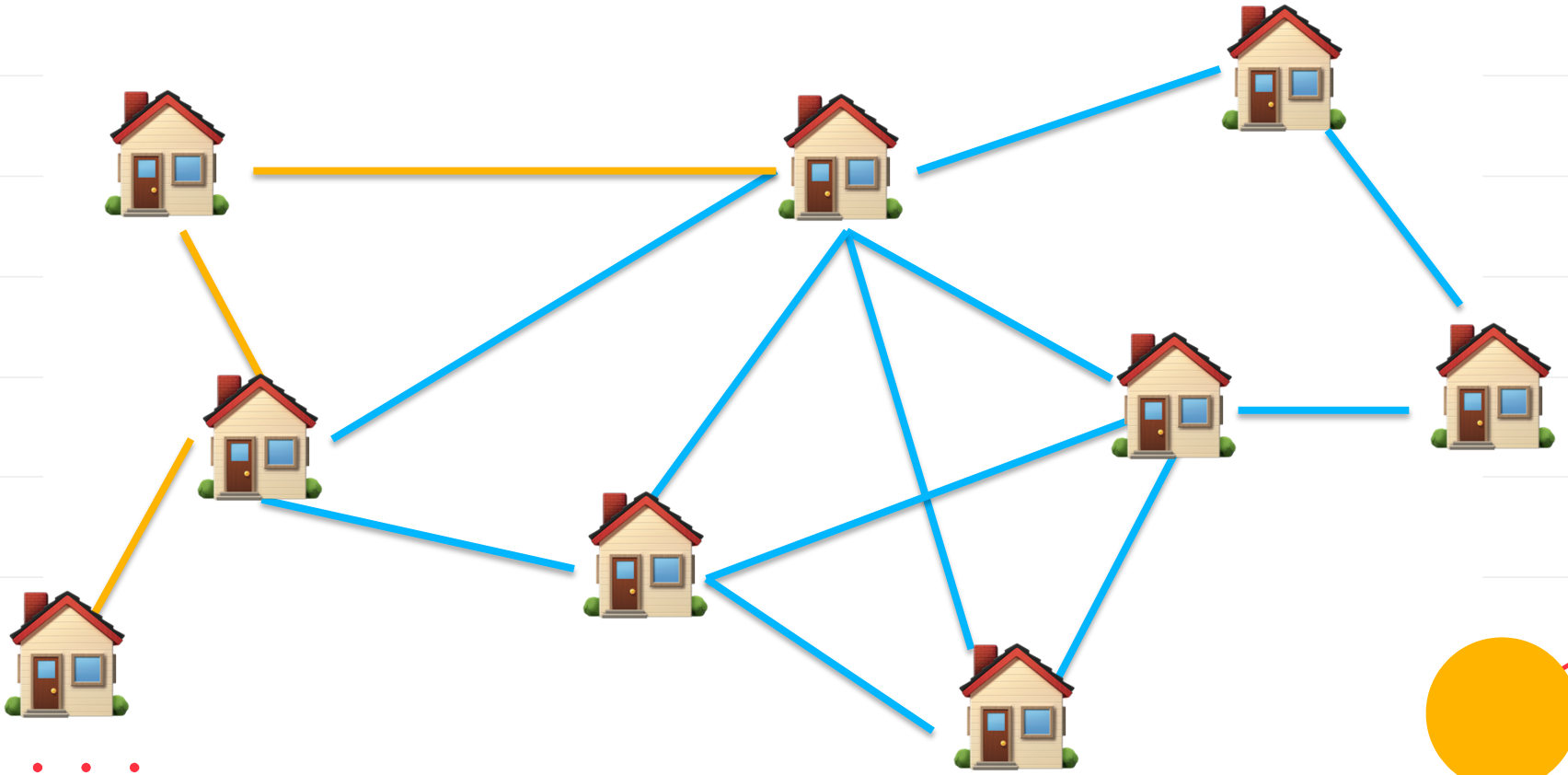
Practice: Hamiltonian Path



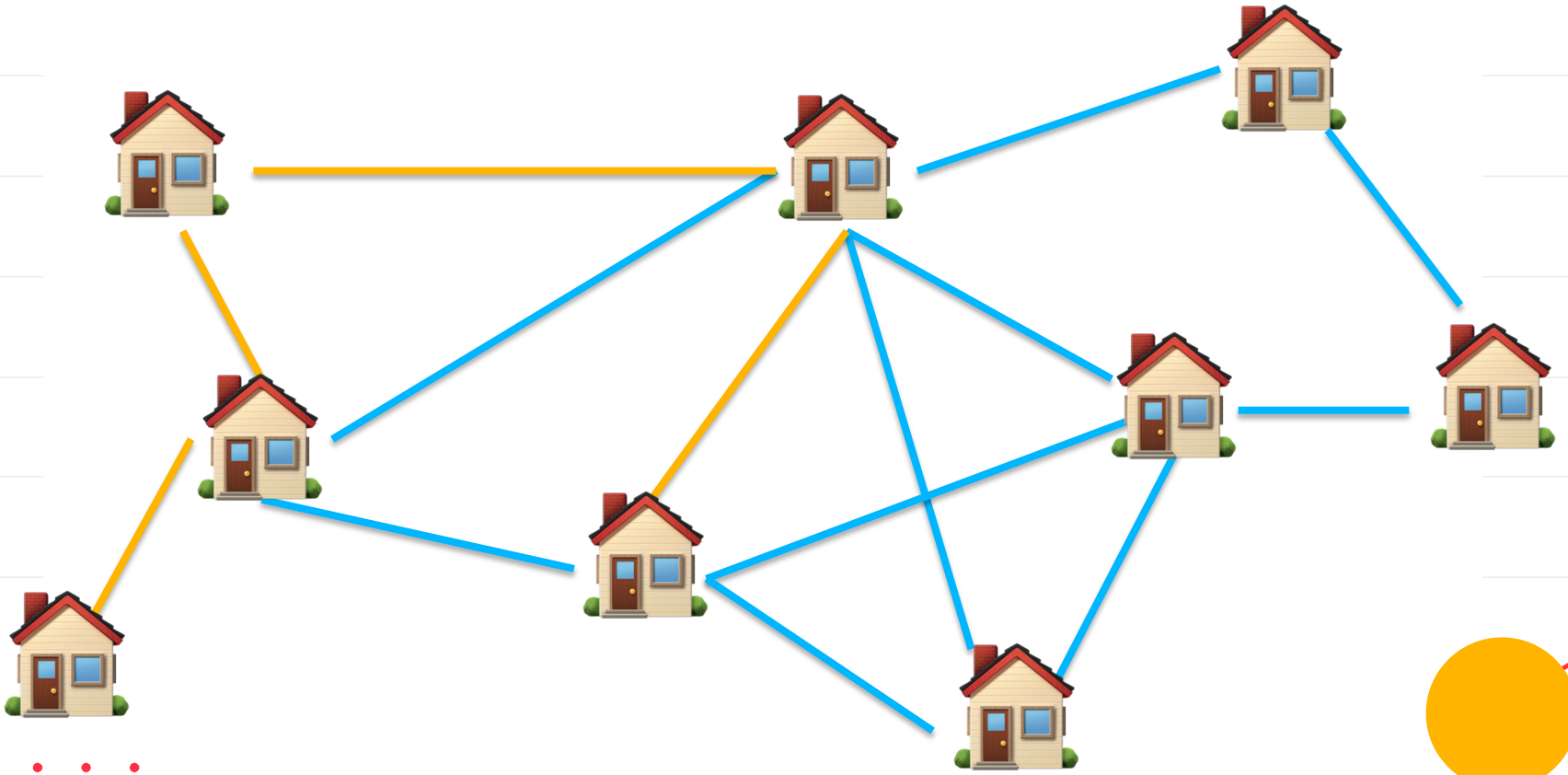
Practice: Hamiltonian Path



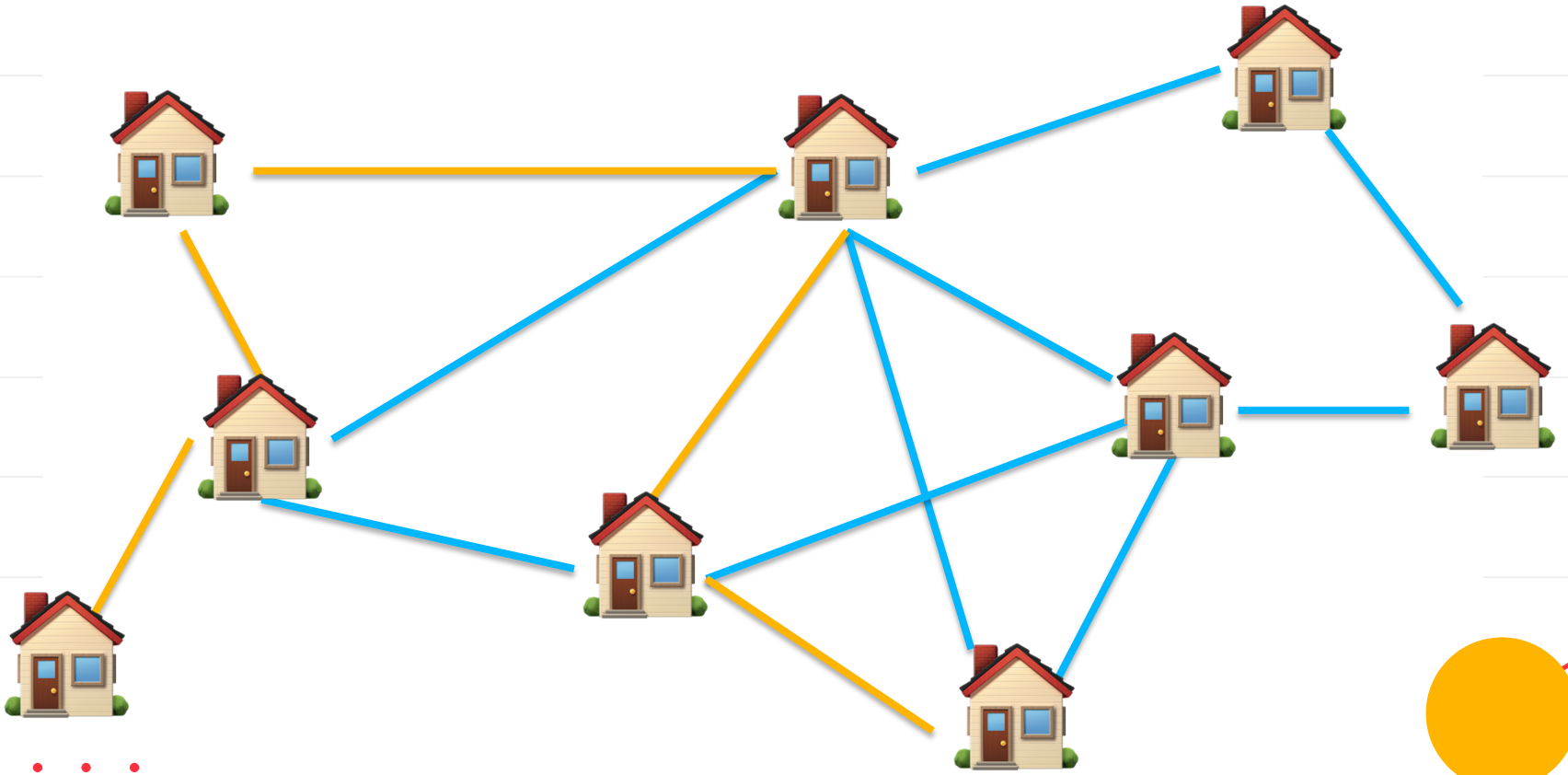
Practice: Hamiltonian Path



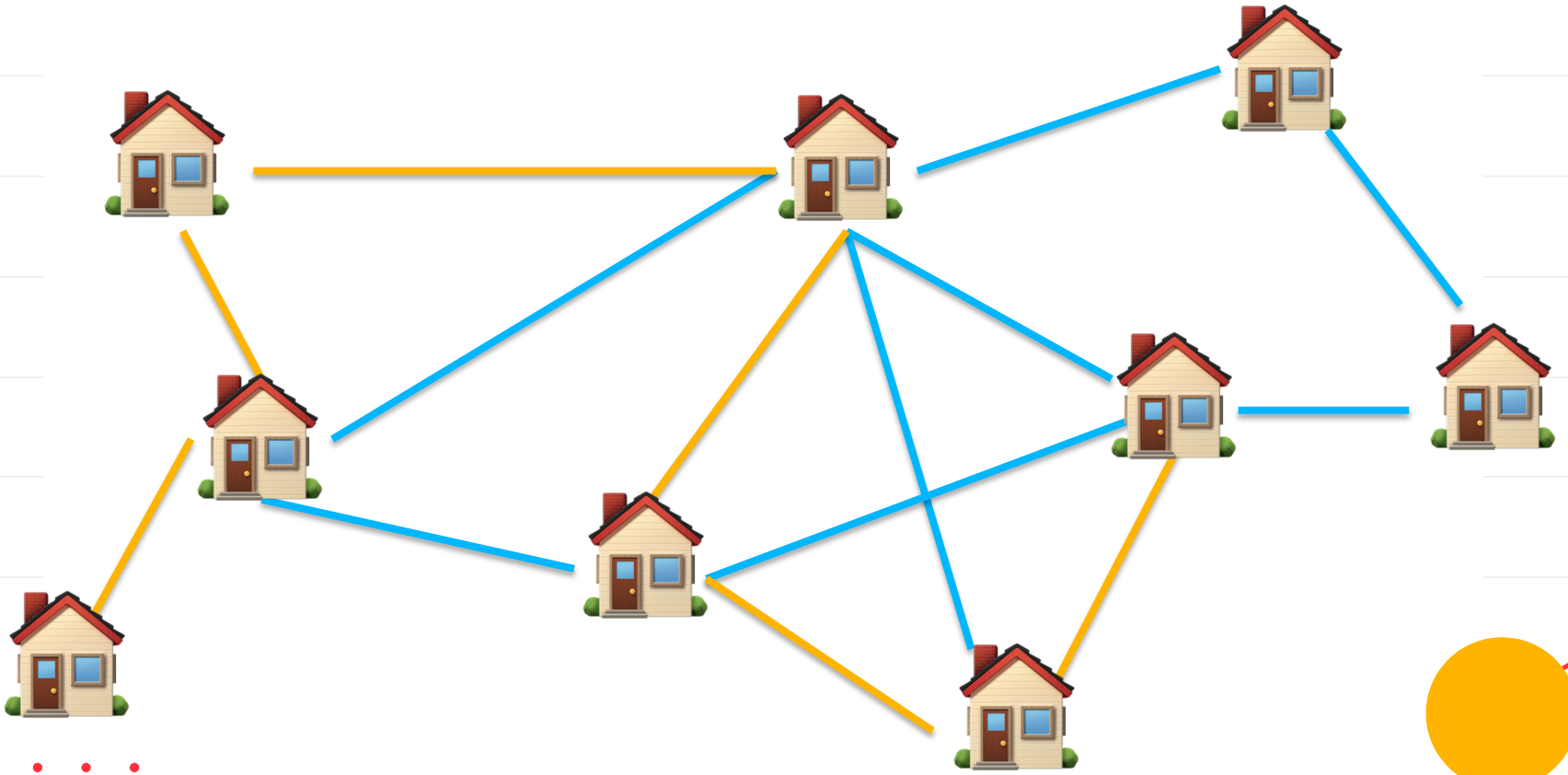
Practice: Hamiltonian Path



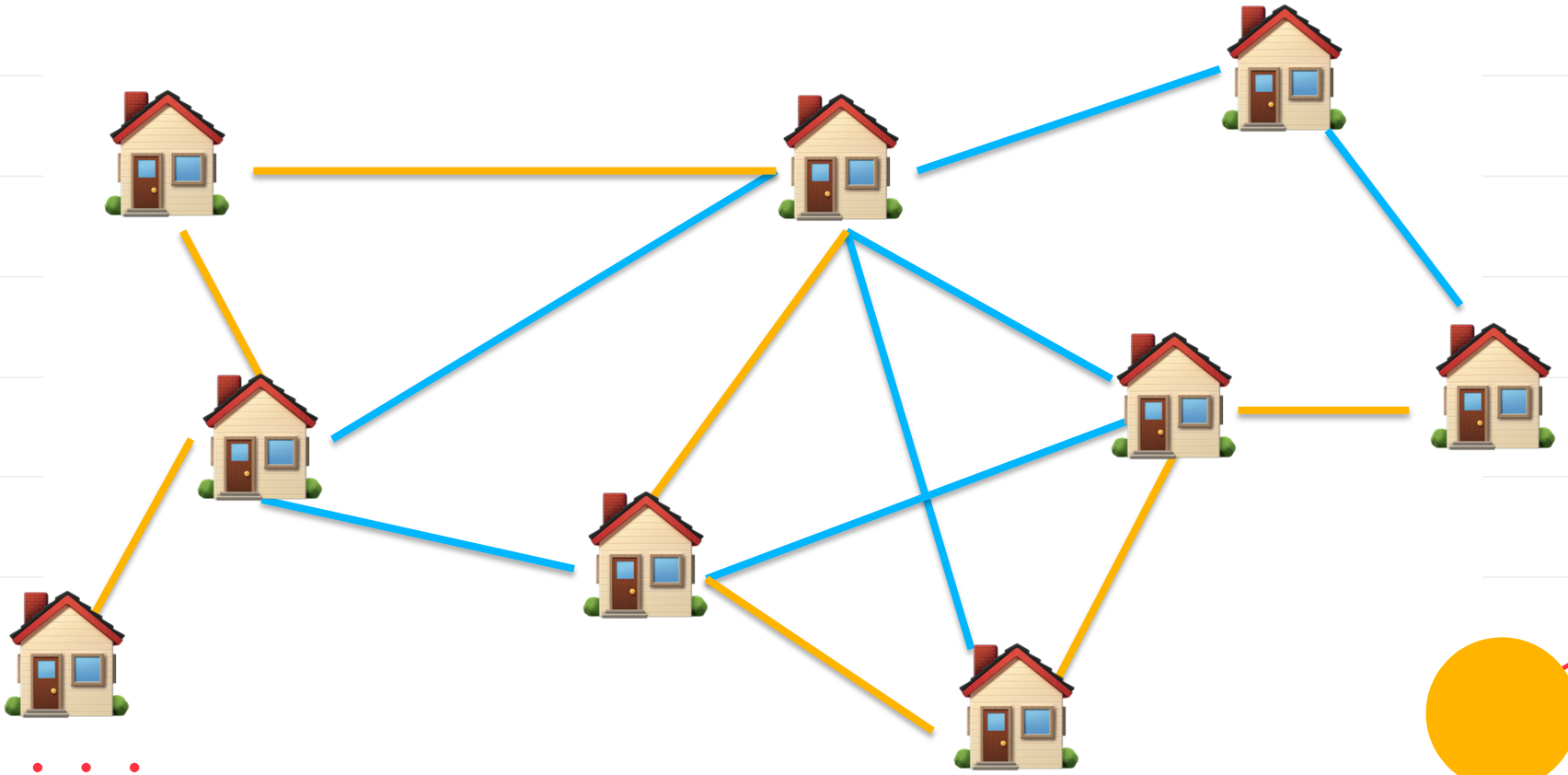
Practice: Hamiltonian Path



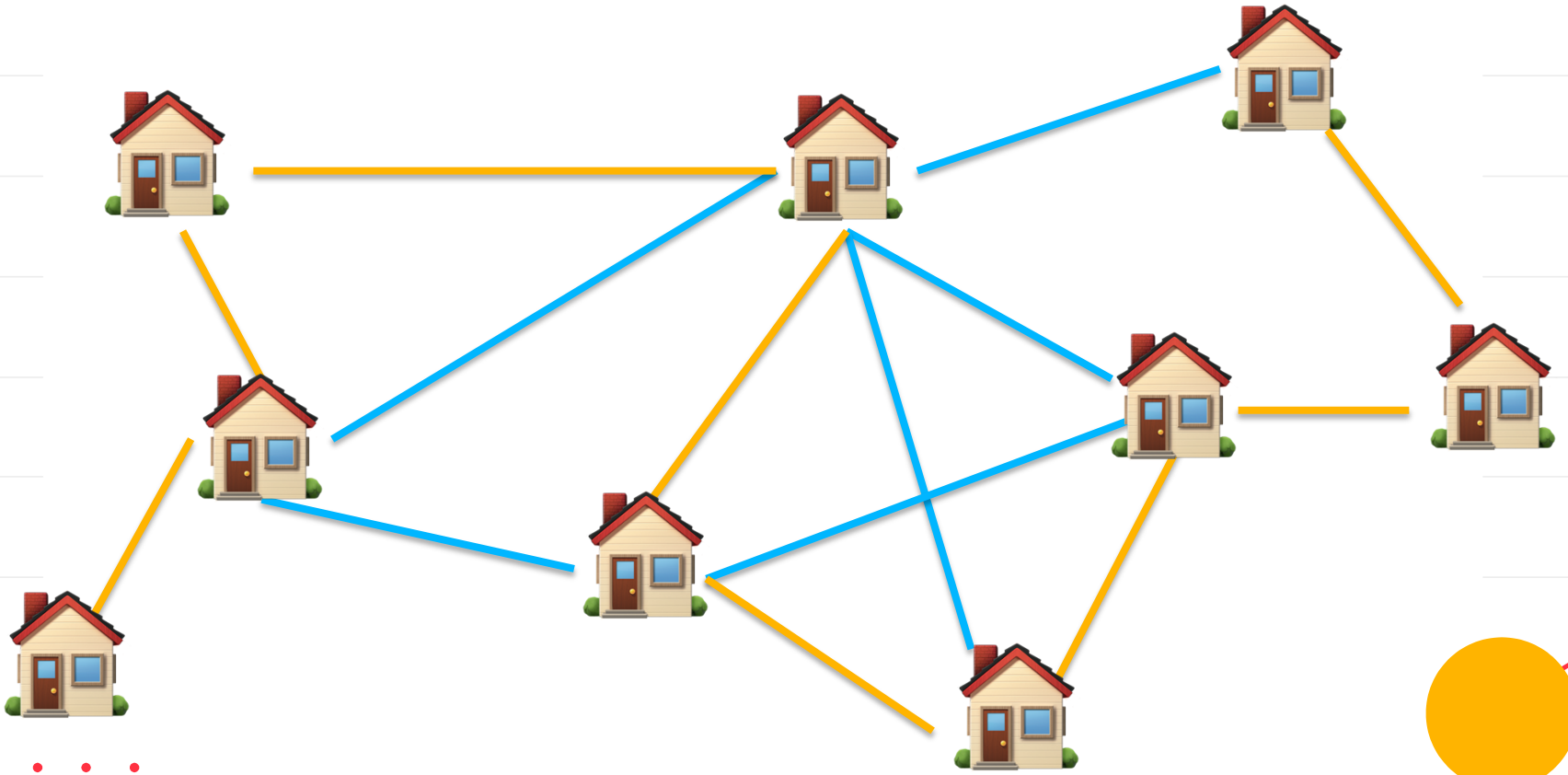
Practice: Hamiltonian Path



Practice: Hamiltonian Path



Practice: Hamiltonian Path



Practice: Hamiltonian Path

We find a transformation witnessing Ham $\leq_p$ SAT

A path has n slots



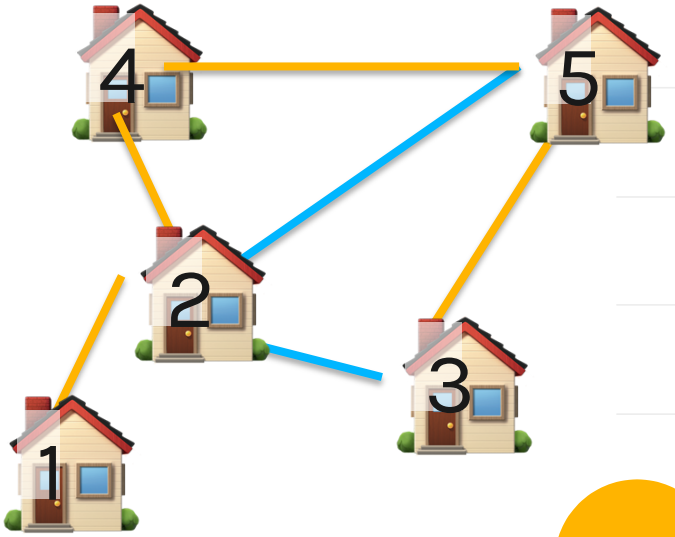
We create n^2 variables x_{ij}.
If x_{ij} is true, node i occupies slot j

Two nodes cannot occupy the same slot

Each slot must be occupied

No node can occupy two different slots

Adjacent slots must be occupied by nodes connected by an edge



Practice: Hamiltonian Path

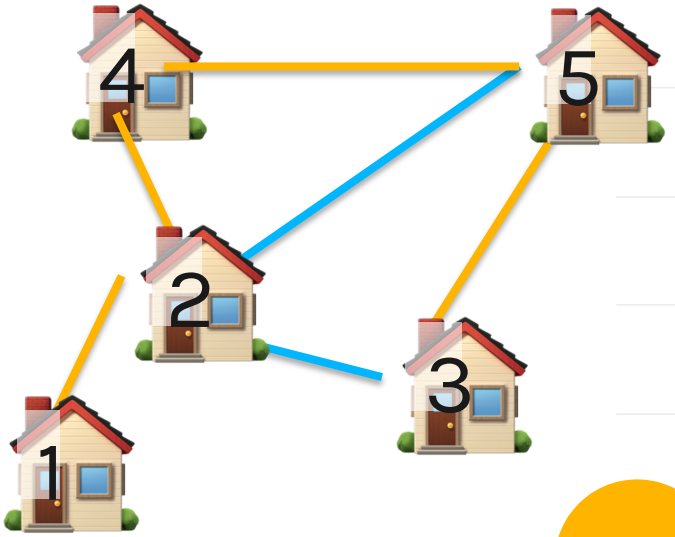
We find a transformation witnessing Ham $\leq_p$ SAT

Two nodes cannot occupy the same slot

$\texttt{exactlyOne}(x_{1k}, x_{2k}, \ldots, x_{nk})$

for each slot k

n^3 clauses

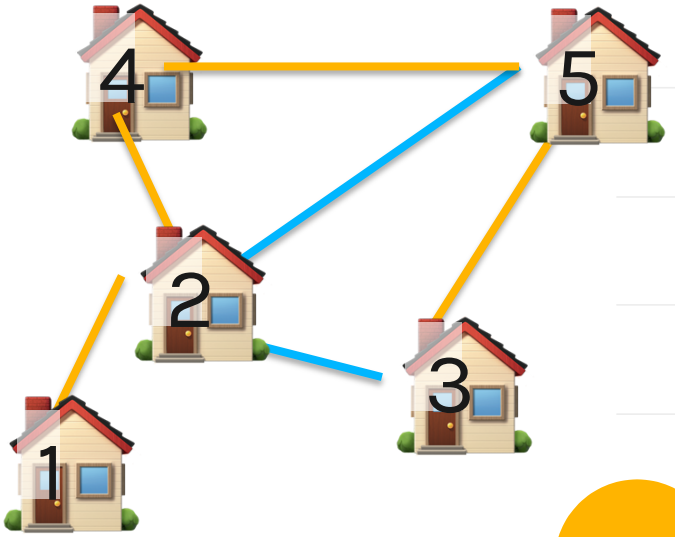


Practice: Hamiltonian Path

We find a transformation witnessing Ham $\leq_p$ SAT

Each slot must be occupied

already covered by the previous clauses



Practice: Hamiltonian Path

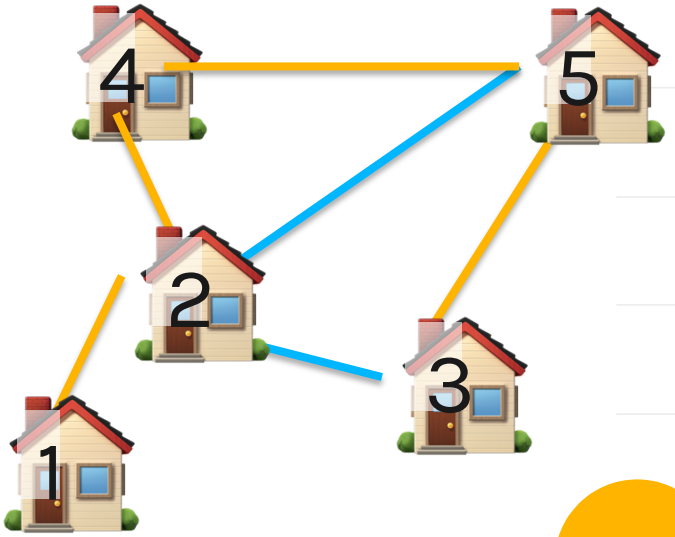
We find a transformation witnessing Ham $\leq_p$ SAT

No node can occupy two different slots

$\texttt{exactlyOne}(x_{i1}, x_{i2}, \ldots, x_{in})$

for each node i

n^3 clauses



Practice: Hamiltonian Path

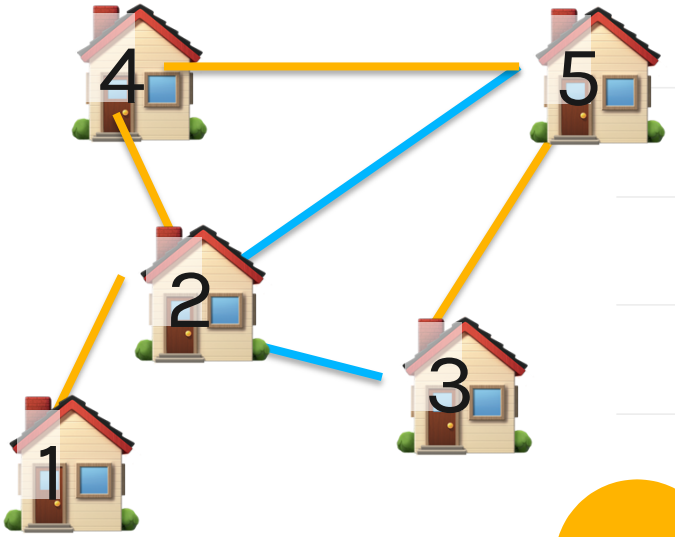
We find a transformation witnessing Ham $\leq_p$ SAT

Adjacent slots must be occupied by nodes connected by an edge



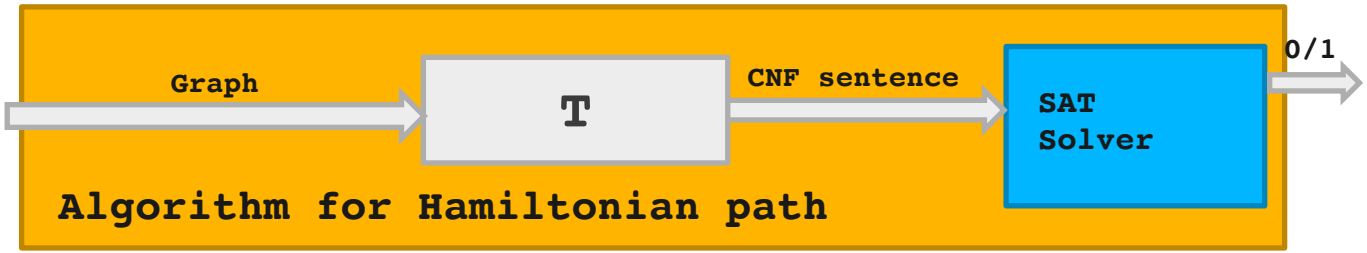
$$\bigwedge_{\substack{(i,j) \notin E \\ k < n}} (\neg x_{ik} \vee \neg x_{jk+1})$$

If no edge between `i` and `j` exists, then some node is not adjacent to the other



Practice: Hamiltonian Path

We must prove that there exists a Hamiltonian path in G ***iff*** the CNF formula we have built from G is satisfiable.



$\text{Ham} \leq_p \text{SAT}$

How is SAT useful to us?

We want to solve a difficult problem f, for which we do not know an algorithm:

We find a transformation witnessing $f \leq_p SAT$
We transform the input to our problem in a SAT formula and run a SAT solver.

usage 1

$LP \leq_p SAT$

$Ham \leq_p SAT$

How is SAT useful to us?

usage 2

We want to solve a difficult problem f, for which we do not know an algorithm:

We want to prove that a problem f, is NP-hard.

We find a transformation witnessing SAT $\leq_p$ f

We can also choose other NP-hard problems, not necessarily SAT.

SAT $\leq_p$ 3-SAT

SAT $\leq_p$ Subset-Sum

SAT $\leq_p$ k-Clique

3-SAT

Is a sentence in CNF, where each clause has exactly 3 literals satisfiable?

SAT $\leq_p$ 3-SAT

For each SAT sentence we build some 3-SAT sentence

The SAT sentence is satisfiable iff the 3-SAT sentence we build is satisfiable

SAT $\leq_p$ 3-SAT

Start off with some SAT sentence:

$$\varphi = C_1 \wedge C_2 \wedge \ldots \wedge C_n$$

If $|C_i| = 1$ $\quad C_i = L$

We invent two variables, and the sentence:

$$D_i = (L \vee x_1 \vee x_2) \wedge (L \vee \neg x_1 \vee x_2) \wedge (L \vee x_1 \vee \neg x_2) \wedge (L \vee \neg x_1 \vee \neg x_2)$$

C_i is satisfiable iff D_i is satisfiable

SAT $\leq_p$ 3-SAT

Start off with some SAT sentence:

$$\varphi = C_1 \wedge C_2 \wedge \ldots \wedge C_n$$

If $|C_i| = 2$ $\quad C_i = L_1 \vee L_2$

We invent one variable, and the sentence:

$$D_i = (L_1 \vee L_2 \vee x) \wedge (L_1 \vee L_2 \vee \neg x)$$

C_i is satisfiable iff D_i is satisfiable

SAT $\leq_p$ 3-SAT

Start off with some SAT sentence:

$\varphi = C_1 \wedge C_2 \wedge \ldots \wedge C_n$

If $|C_i| = 3$ $\quad D_i = C_i$

SAT $\leq_p$ 3-SAT

Start off with some SAT sentence:

$$\varphi = C_1 \wedge C_2 \wedge \ldots \wedge C_n$$

If $|C_i| > 3$ $\quad C_i = L_1 \vee L_2 \vee \ldots \vee L_k$

We invent $k-3$ variables to be used as follows:

$$\begin{aligned} D_i = \; & (L_1 \vee L_2 \vee x_1) \wedge \\ & (\neg x_1 \vee L_3 \vee x_2) \wedge \\ & (\neg x_2 \vee L_4 \vee x_3) \wedge \\ & \ldots \\ & (\neg x_{k-4} \vee L_{k-2} \vee x_{k-3}) \wedge \\ & (\neg x_{k-3} \vee L_{k-1} \vee L_k) \end{aligned}$$

$C_i = L_1 \vee L_2 \vee \ldots \vee L_k$

We prove C_i is satisfiable iff D_i is satisfiable

$$
\begin{aligned}
D_i = {} & (L_1 \vee L_2 \vee x_1) \wedge \\
& (\neg x_1 \vee L_3 \vee x_2) \wedge \\
& (\neg x_2 \vee L_4 \vee x_3) \wedge \\
& \ldots \\
& (\neg x_{k-4} \vee L_{k-2} \vee x_{k-3}) \wedge \\
& (\neg x_{k-3} \vee L_{k-1} \vee L_k)
\end{aligned}
$$

$C_i = L_1 \vee L_2 \vee \ldots \vee L_k$

$$
\begin{aligned}
D = {} & (L_1 \vee L_2 \vee x_1) \wedge \\
& (\neg x_1 \vee L_3 \vee x_2) \wedge \\
& (\neg x_2 \vee L_4 \vee x_3) \wedge \\
& \ldots \\
& (\neg x_{k-4} \vee L_{k-2} \vee x_{k-3}) \wedge \\
& (\neg x_{k-3} \vee L_{k-1} \vee L_k)
\end{aligned}
$$

(⇒) Suppose **C** is satisfiable

Some interpretation **I** makes **C** true.

I makes some L_i true.

$C = L_1 \vee L_2 \vee \ldots L_i \ldots \vee L_k$

$$
\begin{aligned}
D = {} & (L_1 \vee L_2 \vee x_1) \wedge \\
& (\neg x_1 \vee L_3 \vee x_2) \wedge \\
& (\neg x_2 \vee L_4 \vee x_3) \wedge \\
& \ldots \\
& (\neg x_{i-2} \vee L_i \vee x_{i-1}) \wedge \\
& \ldots \\
& (\neg x_{k-4} \vee L_{k-2} \vee x_{k-3}) \wedge \\
& (\neg x_{k-3} \vee L_{k-1} \vee L_k)
\end{aligned}
$$

(⇒) Suppose **C** is satisfiable

Some interpretation **I** makes **C** true.

I makes some L_i true.

$C = L_1 \vee L_2 \vee \dots$

Make x_j true, for all $j \le i-1$

$D = (L_1 \vee L_2 \vee x_1) \wedge$
$(\neg x_1 \vee L_3 \vee x_2) \wedge$
$(\neg x_2 \vee L_4 \vee x_3) \wedge$
$\dots$
$(\neg x_{i-2} \vee L_i \vee x_{i-1}) \wedge$
$\dots$
$(\neg x_{k-4} \vee L_{k-2} \vee x_{k-3}) \wedge$
$(\neg x_{k-3} \vee L_{k-1} \vee L_k)$

This clause is true.

(⇒) Suppose **C** is satisfiable

Some interpretation **I** makes **C** true.

I makes some L_i true.

Make x_j true, for all $j \ge i$

$C = L_1 \vee L_2 \vee \dots L_i \dots \vee L_k$

$$
\begin{aligned}
D = {} & (L_1 \vee L_2 \vee x_1) \wedge \\
& (\neg x_1 \vee L_3 \vee x_2) \wedge \\
& (\neg x_2 \vee L_4 \vee x_3) \wedge \\
& \dots \\
& (\neg x_{i-2} \vee L_i \vee x_{i-1}) \wedge \\
& \dots \\
& (\neg x_{k-4} \vee L_{k-2} \vee x_{k-3}) \wedge \\
& (\neg x_{k-3} \vee L_{k-1} \vee L_k)
\end{aligned}
$$

($\Leftarrow$) Suppose D is satisfiable

If no x_i is true, then $L_1 \vee L_2$ is true.

$C = L_1 \vee L_2 \vee \ldots L_i \ldots \vee L_k$

$$
\begin{aligned}
D = {} & (L_1 \vee L_2 \vee x_1) \wedge \\
& (\neg x_1 \vee L_3 \vee x_2) \wedge \\
& (\neg x_2 \vee L_4 \vee x_3) \wedge \\
& \ldots \\
& (\neg x_{i-2} \vee L_i \vee x_{i-1}) \wedge \\
& \ldots \\
& (\neg x_{k-4} \vee L_{k-2} \vee x_{k-3}) \wedge \\
& (\neg x_{k-3} \vee L_{k-1} \vee L_k)
\end{aligned}
$$

(⇐) Suppose D is satisfiable

If no x_i is false, then $L_{k-1} \vee L_k$ is true.

$C = L_1 \vee L_2 \vee \dots L_i \dots \vee L_k$

$$
\begin{aligned}
D = {} & (L_1 \vee L_2 \vee x_1) \wedge \\
& (\neg x_1 \vee L_3 \vee x_2) \wedge \\
& (\neg x_2 \vee L_4 \vee x_3) \wedge \\
& \dots \\
& (\neg x_{i-2} \vee L_i \vee x_{i-1}) \wedge \\
& \dots \\
& (\neg x_{k-4} \vee L_{k-2} \vee x_{k-3}) \wedge \\
& (\neg x_{k-3} \vee L_{k-1} \vee L_k)
\end{aligned}
$$

(⇐) Suppose D is satisfiable

If x_1 is false then $L_1 \vee L_2$ is true.

$C = L_1 \vee L_2 \vee \dots L_i \dots \vee L_k$

$$
\begin{aligned}
D = {} & (L_1 \vee L_2 \vee x_1) \wedge \\
& (\neg x_1 \vee L_3 \vee x_2) \wedge \\
& (\neg x_2 \vee L_4 \vee x_3) \wedge \\
& \dots \\
& (\neg x_{i-2} \vee L_i \vee x_{i-1}) \wedge \\
& \dots \\
& (\neg x_{k-4} \vee L_{k-2} \vee x_{k-3}) \wedge \\
& (\neg x_{k-3} \vee L_{k-1} \vee L_k)
\end{aligned}
$$

($\Leftarrow$) Suppose D is satisfiable

If some x_i is true and some x_{i+1} is false, then L_{i+2} is true.

If x_{k-3} is true then $L_{k-1} \vee L_k$ is true.

We have proved that 3SAT is NP-complete.